

# Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

*Working draft — not for circulation*

Mainak

**What this document is.** The complete state of the work, including results that are proved and machine-checked but not yet worth putting in the report. Sections 1 to 3 are shared verbatim with `main.tex`; everything after them is parked material. When a parked section earns its place, move its file from `working/` back to `sections/` and uncomment the matching line in `main.tex`. Nothing here is stale: every claim was re-verified against the scripts on the date of its section.

## 1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE<sup>+</sup>16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent  $i$  receives, in addition to her bundle, a subsidy  $p_i \geq 0$  supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with  $m = n$ : under the scaling in which no agent values any single item above 1, a total subsidy of  $n - 1$  suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary  $m$ , gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of  $(n - 1)m$  is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that  $n - 1$  suffices when we may choose the allocation. Brustle et al. [BDN<sup>+</sup>20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations  $2(n - 1)$  per agent suffices, so that the total subsidy is  $O(n^2)$  and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of  $n$  on the dichotomous subclass: when every marginal

$v_i(S \cup \{g\}) - v_i(S)$  lies in  $\{0, 1\}$ , there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most  $n - 1$  in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with  $n$  agents and a single unit-valued good,  $n - 1$  agents must each be paid 1.

**Chores.** Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent  $i$ , every  $S \subseteq M$  and every  $g \in M \setminus S$ ,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}.$$

Equivalently, each marginal  $v_i(S \cup \{g\}) - v_i(S)$  is 0 or  $-1$ : every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost; a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent’s assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

**Why this is not a change of sign.** It is tempting to expect that a chores result follows from the corresponding goods result by negating the valuations. It does not, for three reasons that we record here because they shape everything that follows.

First, money is one-signed. The subsidy vector is constrained to  $p \in \mathbb{R}_+^n$ , and negating an instance negates the valuations but leaves the subsidies where they are. The polyhedron of feasible subsidy vectors is therefore a genuinely different object in the two models, and no formal duality carries a bound across.

Second, the obvious transformation is available only for two agents. Writing  $c_i := -v_i$  for the cost form of a negative dichotomous valuation, the complement  $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$  is again dichotomous in the sense of [BKNS22], so the class is closed under complementation. But the complement transformation respects the allocation only when  $M \setminus A_j = A_i$ , that is,

only when  $n = 2$ ; see Remark 7. For  $n \geq 3$  there is no such reduction, and that is where the content of the problem lies.

Third, and most tellingly, envy flows the other way. Under goods, envy concentrates *into* the agent holding the valuable items: many agents envy one, and the tight instance of [BKNS22] pays  $n - 1$  agents one dollar each so that they stop envying the single agent who received the good. Under chores, envy emanates *out of* the agent carrying the load: one agent envies many. Since the minimum subsidy to an agent is the weight of the heaviest directed path leaving her in the envy graph [HS19, Theorem 2], the two situations are not interchangeable, and one should not assume in advance that the same bound is the right target.

**What is already known, and what is open.** Two facts delimit the problem before any work is done, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the two-sided normalisation  $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ . The reduction of Kawase et al. [KMS<sup>+</sup>24], which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most  $n - 1$  per agent and  $n(n - 1)/2$  in total. [VERIFY: this containment and the resulting bound against Reading\_9 directly; confirm in particular which EF1 convention [KMS<sup>+</sup>24] requires, since the goods-only and two-sided forms of EF1 differ once chores are present]

On the other hand, the lower bound of the goods model survives intact.

**Example 1** (The  $n - 1$  lower bound transfers). Take  $n$  agents and  $m = n - 1$  chores, with  $c_i(S) = |S|$  for every agent  $i$  — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which  $n - 1$  agents hold one chore each and one agent holds nothing, each of the  $n - 1$  loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of  $n - 1$ . No allocation does better: any complete allocation gives some agent a bundle of size  $k \geq 1$  while some other agent is empty, and that agent’s heaviest outgoing path already has weight  $k$ .

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

**Conjecture 2.** *For every instance with negative dichotomous valuations there is an envy-free solution  $(\mathcal{A}, p)$  with  $p \in \{0, 1\}^n$ , hence with total subsidy at most  $n - 1$ ; and it can be computed in polynomial time given value-oracle access.*

Example 1 shows Conjecture 2 would be tight. Between the  $n - 1$  per agent that [KMS<sup>+</sup>24] already gives and the  $\{0, 1\}$  per agent being asked for lies a factor of  $n$  in the total, exactly the gap that [BKNS22] closed on the goods side. Our objective is to close it on the chores side or to exhibit an instance showing that it cannot be closed.

**Scope of this report.** Conjecture 2 is open. This report sets up the problem and stops there: Section 2 fixes the model and notation and records what transfers from the goods

theory unchanged, and Section 3 isolates three structural facts about the chore problem — an arc-update lemma (Lemma 11) reducing the entire goods/chore asymmetry to two lines, a two-tier characterisation of  $\{0, 1\}$  solutions (Proposition 12), and a size-shift transform (Theorem 13) exhibiting the chore problem as exactly the goods problem of [BKNS22] together with a cardinality-balancing requirement. The last of these is the precise sense in which Conjecture 2 does not follow from [BKNS22] as a black box.

Work towards the conjecture is in progress and is deliberately not reported here. Several special cases are settled and two natural proof strategies have been ruled out by explicit machine-checked instances; none of it is stated below until it is worth stating. The remainder of this working draft, from Section 4 onwards, carries that material.

**Related Work.** The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of  $n - 1$ , Goko et al. [GIK<sup>+</sup>21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS<sup>+</sup>24], which is the only work in this line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN<sup>+</sup>20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining  $n - 1$  per agent and  $n(n - 1)/2$  in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for general binary valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. Whether either branch survives the passage to chores is open; this paper addresses the first.

[**TODO:** a literature sweep of the chores-side fair division literature is still outstanding. Nothing in the corpus R1–R9 covers indivisible chores except [KMS<sup>+</sup>24], and the related-work paragraph above therefore says nothing about EF1/EFX for chores, about the goods-and-chores literature, or about prior uses of subsidies for task assignment. Add those readings to References/, extend paper\_map\_R1\_to\_R9.md, and rewrite this paragraph before circulating.]

## 2 Notation and Preliminaries

We study the allocation of  $m$  indivisible items among  $n$  agents, with subsidies. Write  $N = [n]$  for the set of agents and  $M$  for the set of items,  $|M| = m$ . Each agent  $i \in N$  has a valuation  $v_i : 2^M \rightarrow \mathbb{R}$  with  $v_i(\emptyset) = 0$ , and we represent an instance by the tuple  $\langle N, M, \{v_i\}_{i \in N} \rangle$ . Algorithms operate in the standard *value-oracle* model: for each  $v_i$  we assume only an oracle that returns  $v_i(S)$  for a queried set  $S \subseteq M$ , and running time is measured in  $n$ ,  $m$  and the number of oracle calls.

An *allocation*  $\mathcal{A} = (A_1, \dots, A_n)$  is an ordered tuple of pairwise disjoint *bundles*, with  $A_i$  assigned to agent  $i$ . The allocation is *complete* if  $\bigcup_{i \in N} A_i = M$  and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation  $\sigma \in \mathbb{S}_n$  we write  $\mathcal{A}_\sigma := (A_{\sigma(1)}, \dots, A_{\sigma(n)})$  for the allocation in which agent  $i$  receives  $A_{\sigma(i)}$ . The utilitarian welfare of  $\mathcal{A}$  is  $\text{SW}(\mathcal{A}) = \sum_{i \in N} v_i(A_i)$ .

**Definition 3** (Envy-Freeness). *An allocation  $\mathcal{A} = (A_1, \dots, A_n)$  is envy-free (EF) iff  $v_i(A_i) \geq v_i(A_j)$  for all agents  $i, j \in N$ .*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent  $i$  receiving bundle  $A_i$  and subsidy  $p_i$  enjoys utility  $v_i(A_i) + p_i$ , with money entering linearly and at the same exchange rate for every agent.

**Definition 4** (Envy-Free Solution). *An allocation  $\mathcal{A} = (A_1, \dots, A_n)$  and a subsidy vector  $p = (p_1, \dots, p_n) \in \mathbb{R}_+^n$  constitute an envy-free solution  $(\mathcal{A}, p)$  iff  $v_i(A_i) + p_i \geq v_i(A_j) + p_j$  for all  $i, j \in N$ .*

An allocation  $\mathcal{A}$  is *envy-freeable* if some  $p \in \mathbb{R}_+^n$  makes  $(\mathcal{A}, p)$  an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write  $M(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$  for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS<sup>+</sup>24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

**Definition 5** (EF1, two-sided form). *An allocation  $\mathcal{A}$  is envy-free up to one item (EF1) iff for all  $i, j \in N$  there exists  $X \subseteq A_i \cup A_j$  with  $|X| \leq 1$  such that  $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$ .*

### 2.1 Negative Dichotomous Valuations

**Definition 6** (Negative Dichotomous Valuation). *A valuation  $v_i : 2^M \rightarrow \mathbb{R}$  with  $v_i(\emptyset) = 0$  is negative dichotomous iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \quad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

*equivalently iff every marginal  $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$  lies in  $\{-1, 0\}$ .*

Every item is thus a chore for every agent:  $v_i$  is non-increasing, so  $v_i(S) \geq v_i(T)$  whenever  $S \subseteq T$ . We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \quad \text{so} \quad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}.$$

The correspondence  $v_i \leftrightarrow c_i$  is a bijection between negative dichotomous valuations and dichotomous valuations in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement  $c_i(A_i) - p_i \leq c_i(A_j) - p_j$  for all  $i, j$ , read as “my own burden net of my compensation is no worse than what I perceive of yours”. We use whichever of  $v_i$  and  $c_i$  is clearer locally and always state which.

The negation, however, does not carry results across, for the reasons given in Section 1. The following remark isolates the one case in which a transformation does exist, and shows why it is uninformative.

**Remark 7** (Complementation, and why it stops at  $n = 2$ ). For a negative dichotomous  $v_i$  with cost form  $c_i$ , define the *complement*  $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$ . Then  $\tilde{v}_i(\emptyset) = 0$  and, for  $g \notin S$ , writing  $T := M \setminus (S \cup \{g\})$ ,

$$\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) = c_i(M \setminus S) - c_i(T) = c_i(T \cup \{g\}) - c_i(T) \in \{0, 1\},$$

so  $\tilde{v}_i$  is dichotomous in the sense of [BKNS22]. The class is therefore closed under complementation, without any submodularity assumption. Now let  $n = 2$  and let  $\mathcal{A} = (A_1, A_2)$  be complete, so that  $M \setminus A_2 = A_1$  and  $M \setminus A_1 = A_2$ . Then  $-c_i(A_i) = \tilde{v}_i(A_j) - c_i(M)$  and  $-c_i(A_j) = \tilde{v}_i(A_i) - c_i(M)$ , so  $(\mathcal{A}, p)$  is an envy-free solution for the chores instance  $\{v_i\}$  if and only if  $((A_2, A_1), p)$  is an envy-free solution for the goods instance  $\{\tilde{v}_i\}$ : swap the bundles and leave each subsidy attached to its agent. Two-agent chore division is thus exactly two-agent goods division, and [BKNS22] settles it. The step  $M \setminus A_j = A_i$  has no analogue for  $n \geq 3$ , where the complement of one bundle is a union of several others, and no reduction of this kind is available.

## 2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation  $\mathcal{A}$ , the *envy graph*  $G_{\mathcal{A}}$  is the complete weighted directed graph on vertex set  $N$  in which the arc  $(i, j)$  carries weight

$$w_{\mathcal{A}}(i, j) := v_i(A_j) - v_i(A_i) = c_i(A_i) - c_i(A_j),$$

the envy that agent  $i$  has towards agent  $j$ . The weight of a directed path  $P$  is  $w_{\mathcal{A}}(P) = \sum_{(i,j) \in P} w_{\mathcal{A}}(i, j)$ , and  $\ell_{\mathcal{A}}(i)$  denotes the maximum weight of any path in  $G_{\mathcal{A}}$  starting at  $i$ , the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition



on  $v_i$  enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

**Theorem 8** ([HS19]). *For any allocation  $\mathcal{A} = (A_1, \dots, A_n)$ , the following are equivalent.*

- (i)  $\mathcal{A}$  is envy-freeable.
- (ii)  $\mathcal{A}$  maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation  $\sigma$  over  $N$ ,  $\sum_{i \in N} v_i(A_i) \geq \sum_{i \in N} v_i(A_{\sigma(i)})$ .
- (iii) The envy graph  $G_{\mathcal{A}}$  has no positive-weight directed cycle.

**Theorem 9** ([HS19]). *For any envy-freeable allocation  $\mathcal{A}$ , the subsidy vector  $p^*$  given by  $p_i^* := \ell_{\mathcal{A}}(i)$  realises an envy-free solution  $(\mathcal{A}, p^*)$ , and  $p_i^* \leq p_i$  for every  $i \in N$  and every  $p$  such that  $(\mathcal{A}, p)$  is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on  $G_{\mathcal{A}}$ .*

Condition (ii) of Theorem 8 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 9 then charges for it.

Specialising to our model gives the following, which we use throughout.

**Observation 10** (Integrality). *Let  $\{v_i\}_{i \in N}$  be negative dichotomous. Then  $v_i(S) \in \mathbb{Z}_{\leq 0}$  for every  $i$  and every  $S \subseteq M$ ; every arc weight  $w_{\mathcal{A}}(i, j)$  is an integer; and for every envy-freeable allocation  $\mathcal{A}$  the minimum subsidy vector satisfies  $p^* \in \mathbb{Z}_{\geq 0}^n$ .*

**Proof** Fix  $i$  and  $S = \{g_1, \dots, g_k\}$ . Telescoping from  $v_i(\emptyset) = 0$  along any ordering of  $S$  writes  $v_i(S)$  as a sum of  $k$  marginals, each of which lies in  $\{-1, 0\}$  by Definition 6; hence  $v_i(S) \in \mathbb{Z}$  and  $-k \leq v_i(S) \leq 0$ . Consequently each arc weight  $w_{\mathcal{A}}(i, j) = v_i(A_j) - v_i(A_i)$  is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 9,  $p_i^* = \ell_{\mathcal{A}}(i)$  is the maximum such weight over paths starting at  $i$ , and it is non-negative because the empty path has weight 0.  $\square$

Observation 10 is what makes the target statement — a subsidy vector in  $\{0, 1\}^n$  — a statement about a *bound* rather than about rounding: once  $p^*$  is known to be integral, showing  $p_i^* \leq 1$  for every  $i$  is the same as showing  $p^* \in \{0, 1\}^n$ . The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

### 3 Structure of the Chore Problem

Throughout this section we work in cost form, so  $w_{\mathcal{A}}(i, j) = c_i(A_i) - c_i(A_j)$ .

### 3.1 The arc-update table, and the whole of the asymmetry

Every incremental algorithm in this literature adds one item to one bundle and then asks what happened to the envy graph. The answer is two lines long, and comparing those two lines against their goods counterparts accounts for the entire difference between the present problem and that of [BKNS22].

For an allocation  $\mathcal{A}$ , an agent  $x$  and an unallocated chore  $g \notin \bigcup_i A_i$ , write

$$Z^x := (A_1, \dots, A_x \cup \{g\}, \dots, A_n)$$

for the allocation obtained by giving  $g$  to  $x$ , and put

$$\delta_x := c_x(A_x \cup \{g\}) - c_x(A_x), \quad \delta_{i,x} := c_i(A_x \cup \{g\}) - c_i(A_x),$$

both of which lie in  $\{0, 1\}$ . So  $\delta_x$  is what the chore costs  $x$  herself, and  $\delta_{i,x}$  is what it costs in  $i$ 's opinion.

**Lemma 11** (Arc update). *For all  $i, j \in N$ ,*

$$w_{Z^x}(i, j) = w_{\mathcal{A}}(i, j) \quad (i, j \neq x), \quad w_{Z^x}(x, j) = w_{\mathcal{A}}(x, j) + \delta_x, \quad w_{Z^x}(i, x) = w_{\mathcal{A}}(i, x) - \delta_{i,x}.$$

**Proof** Only  $A_x$  changes, so an arc between two agents other than  $x$  has both endpoints untouched. For an arc leaving  $x$ ,  $w_{Z^x}(x, j) = c_x(A_x \cup \{g\}) - c_x(A_j) = w_{\mathcal{A}}(x, j) + \delta_x$ . For an arc entering  $x$ ,  $w_{Z^x}(i, x) = c_i(A_i) - c_i(A_x \cup \{g\}) = w_{\mathcal{A}}(i, x) - \delta_{i,x}$ .  $\square$

The mirror statement for goods is obtained by replacing costs with values: there the arcs *out of* the receiving agent fall and the arcs *into* her rise. The consequences are opposite in a way that matters.

|                          | goods [BKNS22]              | chores (here)               |
|--------------------------|-----------------------------|-----------------------------|
| arcs out of the receiver | fall                        | <b>rise</b> by $\delta_x$   |
| arcs into the receiver   | <b>rise</b>                 | fall by $\delta_{i,x}$      |
| so the receiver must be  | <i>maximally</i> subsidised | <i>minimally</i> subsidised |
| which is guaranteed by   | $M(p) \neq \emptyset$       | $p_x = 0$ for some $x$      |

Both guarantees hold trivially — some agent is always maximally subsidised, and some agent always has  $p_x = 0$  under the minimum subsidy vector. The asymmetry is subtler than the table suggests, and it is what Theorem 22 exploits. Under goods, the harm done by an insertion is confined to paths *ending* at the receiver, and requiring  $x \in M(p)$  caps every such path at  $p_u - p_x \leq 0$  in one stroke. Under chores, the harm is done to paths *leaving* the receiver, and  $p_x = 0$  caps only the paths that *start* at  $x$ . A path that merely passes through  $x$  — entering along an arc from some  $i$  with  $\delta_{i,x} = 0$  and leaving along an arc that has just risen by  $\delta_x = 1$  — is not controlled by any condition on  $p_x$ . That gap is not a defect of the analysis; Section 6 shows it is realised.



### 3.2 The shape of a $\{0, 1\}$ solution

**Proposition 12** (Two-tier characterisation). *Let  $p \in \{0, 1\}^n$  and put  $S := \{i \in N \mid p_i = 1\}$ . Then  $(\mathcal{A}, p)$  is an envy-free solution if and only if*

- (a)  $c_i(A_i) \leq c_i(A_j)$  whenever  $i, j$  lie both in  $S$  or both in  $N \setminus S$ ;
- (b)  $c_i(A_i) \leq c_i(A_j) + 1$  for  $i \in S$  and  $j \in N \setminus S$ ;
- (c)  $c_i(A_i) + 1 \leq c_i(A_j)$  for  $i \in N \setminus S$  and  $j \in S$ .

**Proof** In cost form Definition 4 reads  $c_i(A_i) \leq c_i(A_j) + p_i - p_j$  for all  $i, j$ . Substituting the three possible values 0, 1, -1 of  $p_i - p_j$  gives (a), (b) and (c) respectively.  $\square$

The reading is worth keeping in mind when designing an algorithm. The unpaid agents are exactly envy-free among themselves and *strictly* prefer their own bundle to that of every paid agent, by a full unit; the paid agents are exactly envy-free among themselves and envy-free up to one unit towards the unpaid. This is a certificate format, and it also suggests choosing the partition  $S \mid N \setminus S$  *first* and allocating afterwards, rather than building the allocation up one chore at a time. That is the one place where the incremental template killed in Section 6 is not obviously an obstacle. In the proof of Theorem 16 the set  $S$  turns out to be exactly the agents holding a  $\lceil q/n \rceil$ -share of the universally disliked chores.

### 3.3 The size-shift transform

Remark 7 showed that complementation relates the chore and goods models only for  $n = 2$ . There is a second transform, available for every  $n$ , which does not solve the problem but says precisely how far it is from being a corollary of [BKNS22].

**Theorem 13** (Size shift). *For a negative dichotomous instance with cost functions  $c_i$ , define  $\tilde{v}_i(S) := |S| - c_i(S)$ . Then each  $\tilde{v}_i$  is a dichotomous goods valuation in the sense of [BKNS22], and for every allocation  $\mathcal{A}$  and all  $i, j$ ,*

$$w_{\mathcal{A}}(i, j) = \tilde{w}_{\mathcal{A}}(i, j) + |A_i| - |A_j|,$$

where  $\tilde{w}_{\mathcal{A}}$  denotes the envy graph of the goods instance  $\{\tilde{v}_i\}$ . Consequently every cycle has the same weight in both graphs — so  $\mathcal{A}$  is envy-freeable for the chores  $\{c_i\}$  if and only if it is envy-freeable for the goods  $\{\tilde{v}_i\}$  — while for a path  $P$  from  $u$  to  $v$ ,

$$w_{\mathcal{A}}(P) = \tilde{w}_{\mathcal{A}}(P) + |A_u| - |A_v|, \quad \text{hence} \quad \ell_{\mathcal{A}}(u) = \max_{v \in N} \left[ \tilde{d}_{\mathcal{A}}(u, v) + |A_u| - |A_v| \right],$$

with  $\tilde{d}_{\mathcal{A}}(u, v)$  the maximum weight of a path from  $u$  to  $v$  in the goods graph and  $\tilde{d}_{\mathcal{A}}(u, u) = 0$ .

**Proof**  $\tilde{v}_i(\emptyset) = 0$ , and for  $g \notin S$ ,  $\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) = 1 - (c_i(S \cup \{g\}) - c_i(S)) \in \{0, 1\}$ , so  $\tilde{v}_i$  is dichotomous. For the identity,

$$\tilde{w}_{\mathcal{A}}(i, j) = \tilde{v}_i(A_j) - \tilde{v}_i(A_i) = (|A_j| - c_i(A_j)) - (|A_i| - c_i(A_i)) = w_{\mathcal{A}}(i, j) - |A_i| + |A_j|.$$

Summing along a path  $(i_1, \dots, i_r)$ , the terms  $|A_{i_t}| - |A_{i_{t+1}}|$  telescope to  $|A_{i_1}| - |A_{i_r}|$ ; along a cycle they telescope to 0. The characterisation of envy-freeability by the absence of positive cycles (Theorem 8) then transfers, and the formula for  $\ell_{\mathcal{A}}(u)$  follows from Theorem 9.  $\square$

**Remark 14** (Why Conjecture 2 is not a corollary of [BKNS22]). Theorem 13 says the chore problem is the goods problem of [BKNS22] *plus a cardinality-balancing requirement*. A sufficient condition falls straight out: since  $\tilde{d}_{\mathcal{A}}(u, v) \leq \tilde{p}_u - \tilde{p}_v$  for any envy-eliminating  $\tilde{p}$ , an allocation solving the goods instance in which the quantity  $|A_i| + \tilde{p}_i$  has spread at most 1 across agents satisfies  $\ell_{\mathcal{A}} \leq 1$ . But [BKNS22] offers no control whatever on bundle cardinalities — its output need not even be EF1, by its own Appendix C — so the requirement is not met by invoking it as a black box. Nor can its algorithm simply be re-run with a balancing tie-break, because Theorem 22 rules out the entire item-by-item template on which it is built. The transform also shows the chore problem is at least as expressive as the goods problem, so no easier.

## 4 Our Results

The target is Conjecture 2, stated in Section 1: an envy-free solution with  $p \in \{0, 1\}^n$  and hence total subsidy at most  $n - 1$ . Example 1 supplies the matching lower bound, so it would be tight. By Observation 10 the conjecture is equivalent to the purely graph-theoretic statement that some allocation  $\mathcal{A}$  has no positive-weight cycle in  $G_{\mathcal{A}}$  and satisfies  $\ell_{\mathcal{A}}(i) \leq 1$  for every agent  $i$ .

We do not settle Conjecture 2. What we contribute is a map of the problem: two solved special cases, three structural tools, and — the substance of the paper — two theorems showing that the two most natural proof strategies both fail, one of them the strategy that succeeds on the goods side.

**Solved cases.** Conjecture 2 holds, constructively and in polynomial time, for **binary additive** costs (Theorem 16) and for **identical** costs (Theorem 17). The case  $n = 2$  follows from [BKNS22] through the complementation of Remark 7. The binary additive proof is the one to imitate: dump every chore on somebody who finds it free, then balance whatever nobody wants. Its bound telescopes along paths in the envy graph, which is the mechanism all three tools below try to reproduce.

**Structural tools** (Section 3). The arc-update table of Lemma 11 isolates the entire difference between the goods and chore problems in two lines. Proposition 12 characterises  $\{0, 1\}$  solutions as a two-tier structure, which is the natural certificate format and suggests choosing the tiers before allocating. Theorem 13 exhibits the chore problem as *exactly* the goods problem of [BKNS22] plus a cardinality-balancing requirement, which is the precise sense in which Conjecture 2 is not a corollary of [BKNS22].

**First negative result** (Section 6). Item-by-item insertion — allocate one chore at a time, never move an already-allocated chore, and repair only by permuting whole bundles — is the template of [BKNS22]. Theorem 22 exhibits a three-agent instance on which that template provably cannot be completed: a legal partial state with  $p \in \{0, 1\}^3$  and an unallocated chore such that *every* agent who receives it forces a subsidy of 2, while the full instance admits a zero-subsidy allocation. Any proof of Conjecture 2 must therefore be allowed to re-open

bundles that have already been allocated. We state this bluntly because the template is seductive and patching the insertion invariant is a dead end, not an unsolved exercise.

**Second negative result** (Section 7). The natural repair is to stop building the allocation incrementally and instead select a good allocation globally, the obvious candidate being one that minimises total cost. This suggests the following strengthening.

**Conjecture 15.** *The allocation in Conjecture 2 may in addition be taken utilitarian-optimal, i.e. minimising  $\sum_{i \in N} c_i(A_i)$  over all allocations.*

Proposition 25 refutes Conjecture 15: there is a three-agent, four-chore instance whose *unique* utilitarian-optimal allocation requires a subsidy of 2, although another allocation — of strictly higher total cost — requires no subsidy at all. Utilitarian optimality is therefore not merely an unhelpful restriction but an actively wrong one, and the programme of searching for a tie-breaking rule inside the utilitarian-optimal set is unsound as posed.

**Evidence** (Section 8). Conjecture 2 survives exhaustive verification for  $n = m = 3$  over all 9,880 instances up to agent symmetry, and randomised search over some  $3.5 \times 10^4$  further instances. Section 8 also explains why that evidence is weaker than the counts suggest, and what would strengthen it.

## 5 Solved Cases

Conjecture 2 holds in two restrictions of the model, by explicit polynomial-time constructions. Both proofs run the same way: bound  $w_{\mathcal{A}}(i, j)$  by a difference  $\phi_i - \phi_j$  of a per-agent potential, so that the bound telescopes to  $\phi_{i_1} - \phi_{i_r}$  along a path and to 0 around a cycle. Envy-freeability and the  $\{0, 1\}$  bound then both follow from controlling the spread of  $\phi$ . Reproducing that mechanism in general is the open problem.

### 5.1 Binary additive costs

Let  $c_i(S) = |S \cap D_i|$  for a set  $D_i \subseteq M$  of chores that agent  $i$  dislikes; every other chore is free for her. Call a chore *universally bad* if it lies in  $D_i$  for every  $i$ , write  $U$  for the set of universally bad chores and  $q := |U|$ .

**Theorem 16.** *For binary additive costs, the following allocation is envy-freeable with minimum subsidy  $p^* \in \{0, 1\}^n$ , hence total subsidy at most  $n - 1$ , and it is computable in time  $O(nm)$ .*

*Give each chore  $g \notin U$  to some agent  $i$  with  $g \notin D_i$ ; at least one exists, by definition of  $U$ .  
Distribute  $U$  as evenly as possible, so that  $u_i := |A_i \cap U| \in \{\lfloor q/n \rfloor, \lceil q/n \rceil\}$  for every  $i$ .*

*The bound is tight, by Example 1.*

**Proof** By construction every chore in  $A_i \setminus U$  is free for  $i$ , so  $c_i(A_i) = |A_i \cap D_i| = u_i$ . For any  $j$  we have  $A_j \cap U \subseteq A_j \cap D_i$ , since  $U \subseteq D_i$ , and therefore  $c_i(A_j) = |A_j \cap D_i| \geq u_j$ . Hence

$$w_{\mathcal{A}}(i, j) = c_i(A_i) - c_i(A_j) \leq u_i - u_j.$$

Summing along a path  $P = (i_1, \dots, i_r)$  the right-hand side telescopes:

$$w_{\mathcal{A}}(P) \leq u_{i_1} - u_{i_r} \leq \lceil q/n \rceil - \lfloor q/n \rfloor \leq 1.$$

Around a cycle the same computation gives  $w_{\mathcal{A}}(C) \leq 0$ , so  $G_{\mathcal{A}}$  has no positive-weight cycle and  $\mathcal{A}$  is envy-freeable by Theorem 8. By Theorem 9,  $p_i^* = \ell_{\mathcal{A}}(i) \leq 1$ , and Observation 10 forces  $p^* \in \{0, 1\}^n$ . Since some agent receives  $\lfloor q/n \rfloor$  universally bad chores and hence requires no subsidy, the total is at most  $n - 1$ . Example 1 is binary additive with  $D_i = M$  for every  $i$ , so the bound cannot be improved.  $\square$

The construction is a two-phase version of the shape the general case must reproduce: *dump every chore on somebody who finds it free, then balance whatever nobody wants*. The obstruction to running the same argument for general negative dichotomous costs is that “free for  $i$ ” becomes context dependent. A chore may be free on top of  $A_j$  and cost a full unit on top of  $A_i$ , so there is no fixed set  $D_i$  to compute with and no fixed potential  $u_i$  to telescope against. Theorem 22 shows that this is a genuine obstacle rather than a bookkeeping nuisance.

## 5.2 Identical costs, and two agents

**Theorem 17.** *Suppose  $c_1 = \dots = c_n = c$ . Then every allocation is envy-freeable, and greedily adding each chore to a currently cheapest bundle yields  $p^* \in \{0, 1\}^n$  in time  $O(m(n + \tau))$ , where  $\tau$  is the cost of one oracle call.*

**Proof** For any permutation  $\sigma$  we have  $\sum_i c(A_{\sigma(i)}) = \sum_i c(A_i)$ , since both sides sum the same function over the same multiset of bundles. So condition (ii) of Theorem 8 holds with equality for every allocation, and every allocation is envy-freeable. Moreover  $w_{\mathcal{A}}(i, j) = c(A_i) - c(A_j)$  telescopes *exactly* along a path  $P = (i_1, \dots, i_r)$ , giving  $w_{\mathcal{A}}(P) = c(A_{i_1}) - c(A_{i_r})$  and hence

$$\ell_{\mathcal{A}}(u) = c(A_u) - \min_{j \in N} c(A_j).$$

It therefore suffices to keep  $\max_i c(A_i) - \min_i c(A_i) \leq 1$ . This is an invariant of the greedy rule. It holds vacuously at the empty allocation. Suppose it holds before a step, so that  $c(A_i) \in \{\mu, \mu + 1\}$  for every  $i$ , where  $\mu$  is the current minimum, and let the next chore go to a bundle attaining  $\mu$ . Its cost becomes  $\mu$  or  $\mu + 1$ , because marginals lie in  $\{0, 1\}$ ; every other bundle is unchanged. So all costs still lie in  $\{\mu, \mu + 1\}$  and the invariant survives. The bound  $p^* \in \{0, 1\}^n$  follows with Observation 10.  $\square$

For  $n = 2$  the conjecture holds as well, with total subsidy at most 1: by Remark 7, two-agent chore division is two-agent goods division with the bundles swapped, and [BKNS22, Theorem 4] applies. As noted there, the reduction consumes the identity  $M \setminus A_j = A_i$  and so says nothing for  $n \geq 3$ .

## 6 Approach 1: Item-by-Item Insertion

The algorithm of [BKNS22] maintains an envy-free solution  $(\mathcal{A}^t, p^t)$  over a partial allocation with  $p^t \in \{0, 1\}^n$ , and at each step allocates one further good, repairing the invariant by

permuting whole bundles among the agents when necessary. Nothing already allocated is ever moved between bundles. This section follows that template on the chore side. Two of its steps come out *easier* than in [BKNS22]; the template nevertheless cannot be completed, and Section 6.3 proves it.

Throughout,  $\mathcal{A}$  is a partial allocation,  $g \notin \bigcup_i A_i$  is an unallocated chore, and  $Z^x$ ,  $\delta_x$ ,  $\delta_{i,x}$  are as in Section 3.1.

## 6.1 The easy case is easier than for goods

**Theorem 18** (Free insertion). *Let  $\mathcal{A}$  be envy-freeable and suppose  $\delta_x = 0$  for some agent  $x$ , that is,  $c_x(A_x \cup \{g\}) = c_x(A_x)$ . Then  $Z^x$  is envy-freeable and  $\ell_{Z^x}(i) \leq \ell_{\mathcal{A}}(i)$  for every  $i$ .*

**Proof** By Lemma 11, arcs between agents other than  $x$  are unchanged; arcs out of  $x$  satisfy  $w_{Z^x}(x, j) = w_{\mathcal{A}}(x, j) + \delta_x = w_{\mathcal{A}}(x, j)$ ; and arcs into  $x$  satisfy  $w_{Z^x}(i, x) = w_{\mathcal{A}}(i, x) - \delta_{i,x} \leq w_{\mathcal{A}}(i, x)$ . So every arc weight weakly decreases. No positive-weight cycle can therefore appear, and no path weight can rise; the claims follow from Theorems 8 and 9.  $\square$

It is worth recording what is *absent* from that proof. There is no hypothesis that  $x$  be maximally subsidised, no permutation of bundles, and no extendability machinery: the corresponding step of [BKNS22] needs all three, because for goods the insertion raises the arcs *into* the receiver and those have to be controlled by placing her in  $M(p)$ . Here the insertion lowers them for free. Consequently:

**Observation 19.** *The only case requiring work is a chore whose marginal cost is 1 for every agent on her own current bundle:  $\delta_x = 1$  for all  $x \in N$ . We call this the hard case.*

## 6.2 A cycle-closing bound

The following is the cheapest general handle on  $\ell_{\mathcal{A}}$  available, and it is the chore form of the device [BDN<sup>+</sup>20] uses in its Section 4, where a lower bound of  $-1$  on every arc weight is shown to force a subsidy of at most one per agent.

**Theorem 20** (Cycle-closing bound). *For any envy-freeable allocation  $\mathcal{A}$  and any agent  $u$ ,*

$$\ell_{\mathcal{A}}(u) \leq \max \left( 0, \max_{v \neq u} [c_v(A_u) - c_v(A_v)] \right).$$

**Proof** The empty path contributes 0. Let  $P$  be a nonempty path from  $u$  to some  $v \neq u$ . Appending the arc  $(v, u)$  closes  $P$  into a directed cycle, whose weight is at most 0 because  $\mathcal{A}$  is envy-freeable (Theorem 8). Hence  $w_{\mathcal{A}}(P) \leq -w_{\mathcal{A}}(v, u) = c_v(A_u) - c_v(A_v)$ .  $\square$

**Corollary 21.** *If  $c_v(A_u) \leq c_v(A_v) + 1$  for all  $u, v \in N$  — no agent would suffer more than one extra unit by taking anybody else's bundle — then  $p^* \in \{0, 1\}^n$ .*

The hypothesis is sufficient and far from necessary: an agent with  $c_i \equiv 0$  can absorb everything, which breaks the hypothesis while the subsidy stays 0. It is nevertheless what does the work in Theorem 16, where the telescoping bound  $w_{\mathcal{A}}(i, j) \leq u_i - u_j$  supplies exactly this.

### 6.3 The obstruction

In the hard case, Lemma 11 says every arc leaving the receiving agent  $x$  rises by exactly 1. Since a path starting at  $x$  then rises by 1, the condition  $p_x = 0$  is *necessary*. The following theorem shows that no condition of this kind can be sufficient.

Say that  $(\mathcal{A}, p)$  is *extendable with  $g$*  if there is a permutation  $\sigma$  such that  $\mathcal{A}_\sigma$  is envy-freeable and some agent  $\kappa$  has  $c_\kappa(A_{\sigma(\kappa)} \cup \{g\}) = c_\kappa(A_{\sigma(\kappa)})$  — that is, if after some welfare-preserving reassignment of the existing bundles, somebody can take  $g$  for free and Theorem 18 applies.

**Theorem 22** (The insertion template cannot be completed). *There is a three-agent instance with negative dichotomous valuations, an envy-freeable partial allocation  $\mathcal{A}$  with  $p^* \in \{0, 1\}^3$ , and an unallocated chore  $g$ , such that for every agent  $x \in N$  the allocation  $Z^x$  requires a subsidy of 2 — even though the full instance admits an allocation with subsidy 0.*

**Proof** Take  $M = \{a_1, a_2, g\}$  and

$$c_1(S) = \max(0, |S| - 1), \quad c_2(S) = c_3(S) = |S|.$$

Each is dichotomous:  $c_1$  has marginals 0, 1, 1 as  $|S|$  runs 0, 1, 2, and  $c_2, c_3$  are binary additive. Note that  $c_1$  is *supermodular*, which is legal — neither [BKNS22] nor the present paper restricts to submodular valuations — and the instance genuinely needs it.

Consider the partial allocation  $\mathcal{A} = (\{a_1, a_2\}, \emptyset, \emptyset)$ . Its envy graph has weight matrix

$$(w_{\mathcal{A}}(i, j))_{i,j} = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix},$$

which has no positive-weight cycle, and  $p^* = (1, 0, 0)$ . So  $(\mathcal{A}, p^*)$  is a legal state of the invariant.

The marginal cost of  $g$  is 1 for every agent on her own bundle:  $c_1(\{a_1, a_2, g\}) - c_1(\{a_1, a_2\}) = 2 - 1 = 1$  and  $c_i(\{g\}) - c_i(\emptyset) = 1$  for  $i = 2, 3$ . So we are in the hard case. Moreover  $(\mathcal{A}, p^*)$  is not extendable with  $g$ : agent 1 must keep  $\{a_1, a_2\}$  under any envy-freeable reassignment, since that bundle costs her 1 but costs agents 2 and 3 two apiece, and no reassignment gives anybody a bundle on which  $g$  is free.

Inserting  $g$  into each bundle in turn gives the minimum subsidy vectors

$$Z^1 \mapsto (2, 0, 0), \quad Z^2 \mapsto (2, 1, 0), \quad Z^3 \mapsto (2, 0, 1),$$

each of which has an entry equal to 2. Yet the complete allocation  $(\{a_1\}, \{a_2\}, \{g\})$  has an identically zero envy matrix and subsidy  $(0, 0, 0)$ .  $\square$

The failure is not that envy-freeability breaks — it does not.

**Lemma 23.** *In the hard case, if  $(\mathcal{A}, p^*)$  is not extendable with  $g$ , then  $Z^x$  is envy-freeable for every  $x \in N$ .*

**Proof** Cycles avoiding  $x$  are unchanged by Lemma 11. Let  $C$  be a cycle through  $x$  and let  $i$  be its predecessor of  $x$ . The only arcs of  $C$  that change are the one leaving  $x$ , which rises by  $\delta_x = 1$ , and the one entering  $x$  from  $i$ , which falls by  $\delta_{i,x}$ ; hence  $w_{Z^x}(C) = w_{\mathcal{A}}(C) + 1 - \delta_{i,x}$ .



If  $w_{\mathcal{A}}(C) \leq -1$  then  $w_{Z^x}(C) \leq 0$  immediately. If  $w_{\mathcal{A}}(C) = 0$ , rotating the bundles along  $C$  is a reassignment that preserves total cost, so by Theorem 8(ii) the rotated allocation is again envy-freeable, and in it agent  $i$  receives  $A_x$ . Non-extendability therefore forces  $c_i(A_x \cup \{g\}) \neq c_i(A_x)$ , i.e.  $\delta_{i,x} = 1$ , and again  $w_{Z^x}(C) \leq 0$ . Since  $\mathcal{A}$  is envy-freeable no cycle has  $w_{\mathcal{A}}(C) \geq 1$ , so these cases are exhaustive.  $\square$

So envy-freeability survives every insertion; it is only the  $\{0, 1\}$  bound on the subsidy that breaks. The reason is visible in Section 3.1. A path  $u \rightsquigarrow i \rightarrow x \rightarrow y \rightsquigarrow$  of weight 1 that merely *passes through*  $x$  has its outgoing arc raised by  $\delta_x = 1$  and its incoming arc lowered by  $\delta_{i,x}$ , so it becomes a path of weight 2 whenever  $\delta_{i,x} = 0$ . No condition on  $p_x$  prevents this. In the instance above both  $x = 2$  and  $x = 3$  are hit this way by  $i = 1$ .

Nor can the repair subroutine of [BKNS22] be mirrored. Its navigation rule walks to an agent that has just required a subsidy of at least 2; in the goods world such an agent is automatically maximally subsidised, which is the property the argument needs. In the chore world the same agent automatically has  $p_u = 1$ , and is therefore precisely the kind of agent to whom the chore must *not* be given.

**Remark 24** (What this rules out). Any proof of Conjecture 2 must be allowed to **re-open already-allocated bundles**. The template of [BKNS22] — allocate one item at a time, never move an item that has been allocated, and repair only by permuting whole bundles — is provably insufficient, by Theorem 22. This is a genuine asymmetry between the goods and chore problems and not a gap in the present analysis, so effort spent strengthening the insertion invariant is effort wasted.

## 7 Approach 2: Selecting a Utilitarian-Optimal Allocation

Theorem 22 rules out building the allocation one chore at a time. The natural response is to stop building it at all and instead *select* a good allocation globally, from a set small enough to reason about and rich enough to contain a witness. Utilitarian optimality is the obvious candidate for two reasons. First, by Theorem 8 an allocation minimising  $\sum_i c_i(A_i)$  over *all* allocations, and not merely over reassignments of its own bundles, is automatically envy-freeable, so half of what we need is free. Second, it carries a strong exchange property: if  $g \in A_i$  is *active* for  $i$ , meaning  $c_i(A_i) - c_i(A_i \setminus \{g\}) = 1$ , then  $c_j(A_j \cup \{g\}) = c_j(A_j) + 1$  for every  $j$  — nobody can absorb it for free, or the allocation would not have been optimal. Combined with Theorem 20, a path of weight at least 2 from  $u$  to  $v$  forces  $c_v(A_u) \geq c_v(A_v) + 2$ , so  $A_u$  carries at least two units that are live for  $v$ , and one would like to transfer one of them along the path and show a potential drops.

This is what Conjecture 15 proposes. It is false.

### 7.1 Utilitarian optimality is the wrong restriction

**Proposition 25.** *Conjecture 15 is false. There is an instance with  $n = 3$  agents and  $m = 4$  chores whose unique utilitarian-optimal allocation requires a subsidy of 2, although another allocation — of strictly greater total cost — admits a subsidy of 0.*

**Proof** Take  $M = \{g_1, g_2, g_3, g_4\}$  and the costs tabulated below.

| $S$         | $c_1$ | $c_2$ | $c_3$ | $S$            | $c_1$ | $c_2$ | $c_3$ |
|-------------|-------|-------|-------|----------------|-------|-------|-------|
| $\emptyset$ | 0     | 0     | 0     | $g_3g_4$       | 2     | 1     | 1     |
| $g_1$       | 1     | 1     | 1     | $g_1g_2g_3$    | 2     | 3     | 2     |
| $g_2$       | 1     | 1     | 1     | $g_1g_2g_4$    | 3     | 2     | 2     |
| $g_3$       | 1     | 1     | 0     | $g_1g_3g_4$    | 2     | 2     | 2     |
| $g_4$       | 1     | 1     | 1     | $g_2g_3g_4$    | 2     | 2     | 2     |
| $g_1g_2$    | 2     | 2     | 2     | $g_1g_2g_3g_4$ | 3     | 3     | 3     |
| $g_1g_3$    | 1     | 2     | 1     |                |       |       |       |
| $g_1g_4$    | 2     | 2     | 2     |                |       |       |       |
| $g_2g_3$    | 1     | 2     | 1     |                |       |       |       |
| $g_2g_4$    | 2     | 2     | 2     |                |       |       |       |

Each column is 0 at  $\emptyset$  and has every marginal in  $\{0, 1\}$ , so all three are negative dichotomous in cost form.

*The utilitarian optimum.* Agent 1 pays 1 for every singleton, so  $c_1(A_1) = 0$  only if  $A_1 = \emptyset$ ; agent 3 pays 0 for  $\{g_3\}$  but 1 for every superset, so  $c_3(A_3) = 0$  only if  $A_3 \subseteq \{g_3\}$ . Any allocation of total cost below 2 would need at least two agents at cost 0, forcing  $A_1 = \emptyset$  and  $A_3 \subseteq \{g_3\}$  and hence  $A_2 \supseteq \{g_1, g_2, g_4\}$ , which costs agent 2 at least 2. So the minimum total cost is 2, attained only by

$$\mathcal{A}^* = (\emptyset, \{g_1, g_2, g_4\}, \{g_3\}), \quad 0 + 2 + 0 = 2.$$

Its envy graph has  $w_{\mathcal{A}^*}(2, 1) = 2 - 0 = 2$ , so  $\ell_{\mathcal{A}^*}(2) \geq 2$ ; in fact the minimum subsidy is  $p^* = (0, 2, 0)$ .

*A zero-subsidy allocation.* Take  $\mathcal{B} = (\{g_1, g_3\}, \{g_2\}, \{g_4\})$ , of total cost  $1 + 1 + 1 = 3$ . Its envy graph has weights

$$(w_{\mathcal{B}}(i, j))_{i,j} = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

every entry of which is at most 0. Hence there is no positive-weight cycle, every directed path has weight at most 0, and  $p^* = (0, 0, 0)$ .  $\square$

Two features make this sharper than a bare counterexample. The utilitarian-optimal allocation is *unique*, so there is no tie to break and no selection rule inside the optimal set can rescue the conjecture. And the gap is the largest possible in one step: subsidy 2 against subsidy 0, at a cost of one extra unit of total burden. Restricting attention to utilitarian-optimal allocations is therefore not a neutral simplification but an actively wrong one.

**Remark 26** (What this rules out). Conjecture 2, if true, is *not* witnessed by welfare-maximising allocations, and the programme of finding a tie-breaking rule inside the utilitarian-optimal set is unsound as posed — on this instance the set is a singleton and the witness lies outside it. The exchange property quoted at the head of this section remains true and remains usable; what fails is the assumption that a witness may be sought among cost-minimising allocations at all. Any search procedure must be free to give up total cost.

## 7.2 Selection rules that fail

Before Proposition 25 was found, three selection rules inside the utilitarian-optimal set were tested, and all three fail on instances where some optimal allocation *does* work. They are recorded because each is a natural first guess.

- **Minimising total perceived load**  $\Psi(\mathcal{A}) := \sum_i \sum_j c_j(A_i)$ , which prefers bundles that are cheap in everybody’s opinion rather than only in their holder’s.
- **Leximin on the perceived-load vector**  $(\max_j c_j(A_i))_{i \in N}$ , which tries to prevent any single bundle from looking heavy to anyone.
- **Local search by single chore transfers** on the objective  $(\max_i \ell_{\mathcal{A}}(i), \sum_i \ell_{\mathcal{A}}(i))$  restricted to the utilitarian-optimal set. This has bad local minima: there are optimal allocations with subsidy 2 from which no single transfer that stays optimal improves the objective.

The local-search failure is the informative one. Single transfers are too coarse a move set, because the utilitarian-optimal allocations of a dichotomous instance are the bases of a structure whose natural moves are exchange *paths and cycles*, not single steps — the same phenomenon that makes Yankee Swap and matroid union the right tools in the matroid-rank literature [BEF21]. A move set matched to that structure is the obvious thing to try next, and remains untried. [TODO: implement exchange-path local search over the full allocation lattice — not restricted to the utilitarian-optimal set, per Remark 26 — and re-run the counterexample hunt with it.]

## 8 Computational Evidence

All searches operate directly on the graph-theoretic form of Conjecture 2: enumerate allocations, build the envy graph, detect positive-weight cycles and compute  $\ell_{\mathcal{A}}$  by a Bellman–Ford iteration on maximum-weight walks, and record  $\min_{\mathcal{A}} \max_i \ell_{\mathcal{A}}(i)$ .

| Experiment                                                                                     | Script     | Result                                                                                            |
|------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------|
| Exhaustive: $n = m = 3$ , all 9,880 instances up to agent symmetry                             | search.py  | worst-case minimum per-agent subsidy = 1                                                          |
| Randomised: $n \in \{3, 4, 5\}$ , $m \in \{4, 5, 6\}$ , $\approx 3.5 \times 10^4$ instances    | fast.py    | no counterexample                                                                                 |
| Binary additive construction of Theorem 16, $2 \times 10^5$ instances, $n \leq 5$ , $m \leq 7$ | binadd.py  | maximum per-agent subsidy ever observed = 1                                                       |
| The obstruction of Theorem 22                                                                  | deadend.py | all three insertions require subsidy 2; full instance solvable with 0                             |
| Does <i>some</i> utilitarian-optimal allocation attain $\leq 1$ ?                              | msw.py     | yes, in all but one of $\approx 2.6 \times 10^4$ instances                                        |
| Does <i>every</i> utilitarian-optimal allocation attain $\leq 1$ ?                             | msw.py     | no — fails in $\approx 2.5\%$ of instances at $n = 3, m = 4$ , rising to $\approx 5\%$ at $m = 5$ |
| The single exception                                                                           | mswcex.py  | refutes Conjecture 15 (Proposition 25)                                                            |

The exhaustive row is the only unconditional evidence for Conjecture 2. The dichotomous functions on  $m$  items number 2, 6, 38, 990 for  $m = 1, 2, 3, 4$ . Agents are interchangeable, so an instance is a multiset of three of them, and at  $m = 3$  there are  $\binom{38+3-1}{3} = \binom{40}{3} = 9,880$  of these — feasible to enumerate, and already out of reach at  $m = 4$ .

The last three rows are the ones that changed the shape of the problem. The utilitarian-optimal set almost always contains a witness, but not always, and one exception is enough: Proposition 25 is built from the single instance in which it did not.

## 8.1 Why the randomised evidence is weak

The count of  $\approx 3.5 \times 10^4$  instances without a counterexample overstates the support for Conjecture 2, and the reason is structural rather than a matter of sample size.

The generator builds a dichotomous function by visiting subsets in order of increasing size and, at each subset  $S$ , computing the interval  $[\ell, h]$  of values consistent with what has already been fixed — namely  $\ell = \max_{g \in S} c(S \setminus \{g\})$  and  $h = \min_{g \in S} c(S \setminus \{g\}) + 1$  — then choosing an endpoint by an unbiased coin whenever  $\ell \neq h$ . Independent fair coins put almost all of the mass on functions that wander in the middle of the feasible region. Structured extremal functions, which take the same endpoint at every subset, are exponentially unlikely: on  $m$  items there are  $2^m - 1$  nonempty subsets and a function that requires a prescribed endpoint at each of them is sampled with probability  $2^{-(2^m-1)}$ , which is  $1/128$  at  $m = 3$  and about  $3 \times 10^{-5}$  at  $m = 4$ , per agent.

This matters because the two instances that carry the results of this paper are of exactly that kind. The obstruction of Theorem 22 uses  $c_1(S) = \max(0, |S| - 1)$  alongside  $c_2 = c_3 = |S|$ , all three of which are endpoint-constant; drawing that triple at  $m = 3$  has probability  $2^{-21} \approx 5 \times 10^{-7}$ , so a run of  $3.5 \times 10^4$  samples would be expected to produce it about once in every sixty runs. Both instances were found by hand or by a targeted search, not by the sampler. The sampler has, in effect, never looked in the region where the interesting behaviour lives.

**Remark 27** (The experiment worth running). If Conjecture 2 is false, the counterexample is in the corner the current generator cannot reach, and no amount of additional uniform sampling will find it. An adversarial generator — biased towards endpoint-constant functions, towards supermodular costs, towards agents whose free chores overlap heavily, and towards instances in which the utilitarian optimum is unique — is the single highest-value experiment outstanding. It should be built before any further effort is spent on proof attempts, because it is the only cheap way to learn whether there is a theorem to prove.

## 8.2 Reproducing the results

| Script         | Purpose                                          |
|----------------|--------------------------------------------------|
| gen.py         | enumerate all dichotomous functions on $m$ items |
| search.py      | exhaustive search, arguments $m$ $n$             |
| fast.py        | randomised counterexample hunt, $m$ $n$ $T$ seed |
| msw.py         | utilitarian-optimal test, $m$ $n$ $T$ seed       |
| tiebreak.py    | selection rules of Section 7.2                   |
| localsearch.py | local minima of single-transfer search           |
| deadend.py     | Theorem 22                                       |
| binadd.py      | Theorem 16                                       |
| mswcex.py      | Proposition 25                                   |

The instance of Proposition 25 was produced by `msw.py 4 3 3000 11` at trial 2460 and is re-derived independently by `mswcex.py`, which re-checks that all three cost functions are negative dichotomous, that the utilitarian optimum is unique, and that a zero-subsidy allocation exists outside it.

## 9 Ideas and Open Threads

A running log, newest last. Nothing in this section is claimed to be true.

### 2026-08-01 — What the two negative results jointly require

Taking Theorem 22 and Proposition 25 together, any correct algorithm must be free to (i) move already-allocated chores between bundles and (ii) give up total cost while doing so. Every construction that currently works does so by exhibiting a per-agent potential  $\phi$  with  $w_{\mathcal{A}}(i, j) \leq \phi_i - \phi_j$ , which telescopes along paths and vanishes around cycles. *Finding such a potential for general negative dichotomous costs, or proving that none exists, is the problem.* In Theorem 16 the potential is  $\phi_i = u_i$ , the count of universally disliked chores held; in Theorem 17 it is  $\phi_i = c(A_i)$ . What is the right  $\phi$  in general?

Worth noting that the existence of such a  $\phi$  is not merely sufficient but close to necessary: if  $p^* \in \{0, 1\}^n$  then  $\phi_i := p_i^*$  satisfies  $w_{\mathcal{A}}(i, j) \leq \phi_i - \phi_j$  by definition of an envy-free solution. So the search for a potential and the search for a proof are the same search, and the content is in constructing  $\phi$  and the allocation *together*.

### Template for the next entry

[**TODO:** replace this with the next idea. Keep the three-part shape: what the idea is, why it might work, what would kill it.]

## 10 Conclusion and Future Work

We have not settled Conjecture 2, and the honest summary of this paper is that it makes the conjecture harder to prove than it looked. Two strategies that a reader would reasonably try

first are ruled out: item-by-item insertion, which is the strategy that works in [BKNS22] (Theorem 22), and restriction to utilitarian-optimal allocations (Proposition 25). What remains is a small set of positive results — binary additive costs (Theorem 16), identical costs (Theorem 17), two agents — and three structural tools that between them say precisely how the chore problem differs from its goods mirror.

The single most useful thing to keep in mind is the constraint the two negative results jointly impose: *a correct algorithm must be free to move already-allocated chores between bundles, and free to give up total cost while doing so*. Every construction in this paper that succeeds does so by exhibiting a per-agent potential  $\phi$  with  $w_{\mathcal{A}}(i, j) \leq \phi_i - \phi_j$ , which telescopes along paths. Finding such a potential for general negative dichotomous costs, or proving none exists, is the problem.

### Directions.

1. **Adversarial instance generation** (Section 8.1). The randomised evidence is drawn from a distribution that essentially never produces the structured extremal functions from which both negative results are built. This is cheap, and it is the only inexpensive way to find out whether Conjecture 2 is true before investing in a proof.
2. **Exchange paths rather than single transfers** (Section 7.2). Single-chore local search has bad local minima. The move set should match the exchange structure of dichotomous costs, as Yankee Swap and matroid union do in the matroid-rank literature [BEF21]; and by Remark 26 the search must range over all allocations, not the utilitarian-optimal ones.
3. **Build the two tiers first** (Proposition 12). Choose the partition into paid and unpaid agents and only then allocate. This is the one natural approach that Theorem 22 does not obviously kill, since it is not incremental in the chores. In the proof of Theorem 16 the paid set is exactly the agents holding a  $\lceil q/n \rceil$ -share of the universally disliked chores; identifying what plays that role in general is the first question.
4. **Climb the milestone ladder**. Binary additive is done (Theorem 16). Binary submodular is the next rung: for matroid-rank costs, matroid union should supply the cardinality balancing that Theorem 13 identifies as the missing ingredient. General dichotomous is the summit.
5. **The mixed model**. Allow each item to be a good or a chore for each agent, with all marginals in  $\{-1, 0, 1\}$ : the dichotomous restriction of the doubly monotone setting of [KMS<sup>+</sup>24]. This is the natural common generalisation of [BKNS22] and of the present paper.

**Before investing further.** The best published bound covering negative dichotomous valuations is that of [KMS<sup>+</sup>24]:  $n - 1$  per agent and  $n(n - 1)/2$  in total, from any EF1 allocation. Conjecture 2 would improve the total by a factor of  $n$  on this subclass — the same factor [BKNS22] gains over [BDN<sup>+</sup>20] on the goods side. [TODO: confirm that nobody has published “dichotomous chores with subsidy” already. This is the one check that could make the whole programme redundant and it has not been done.]



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